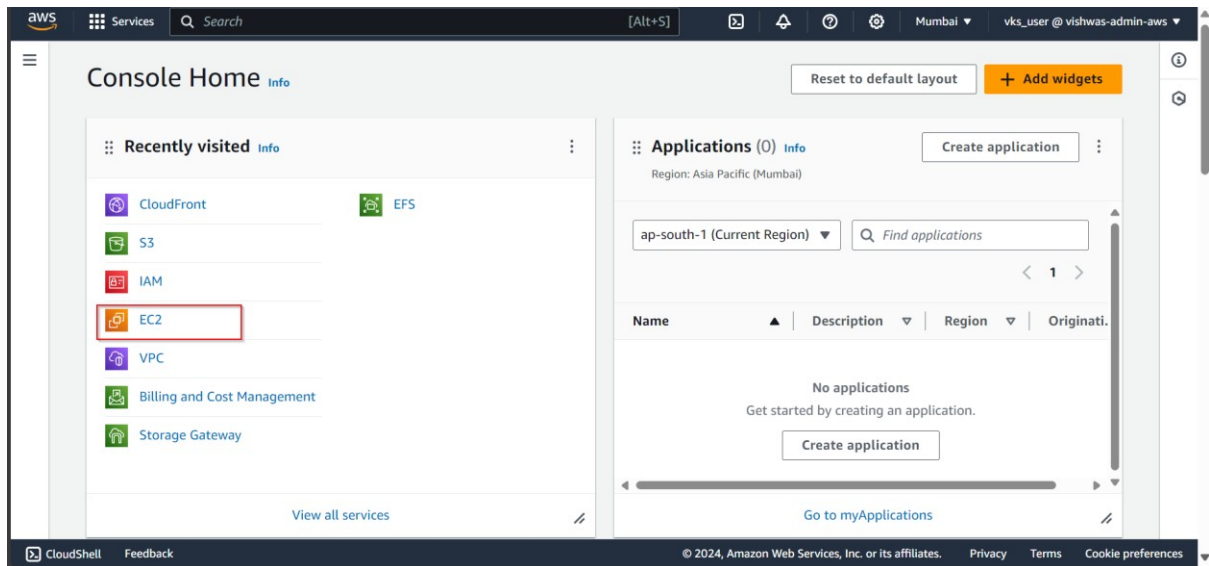
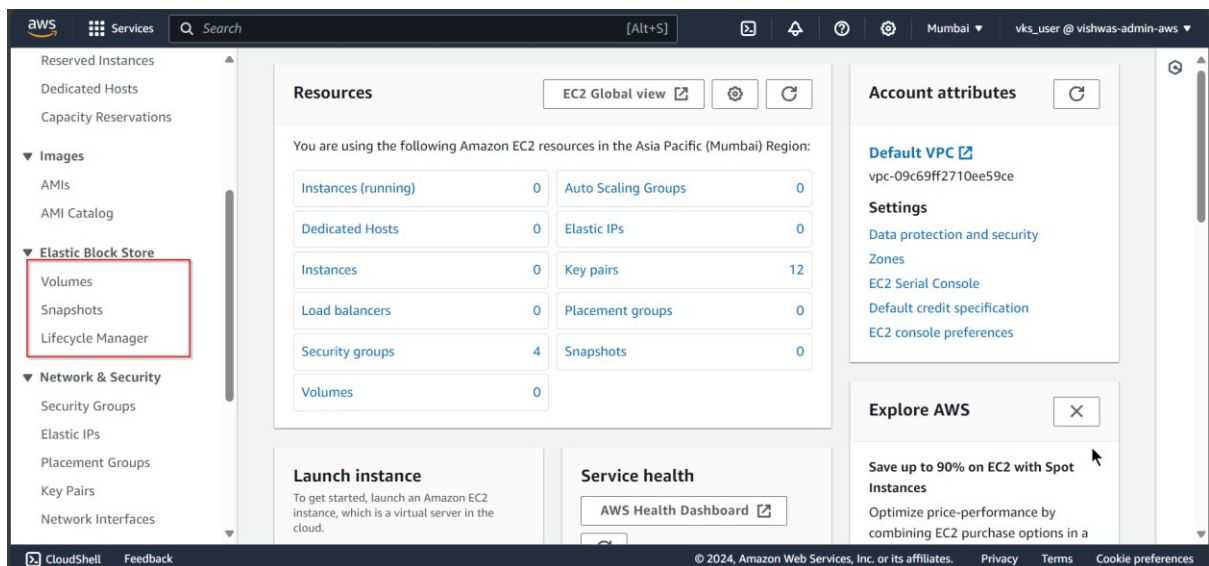


EBS :

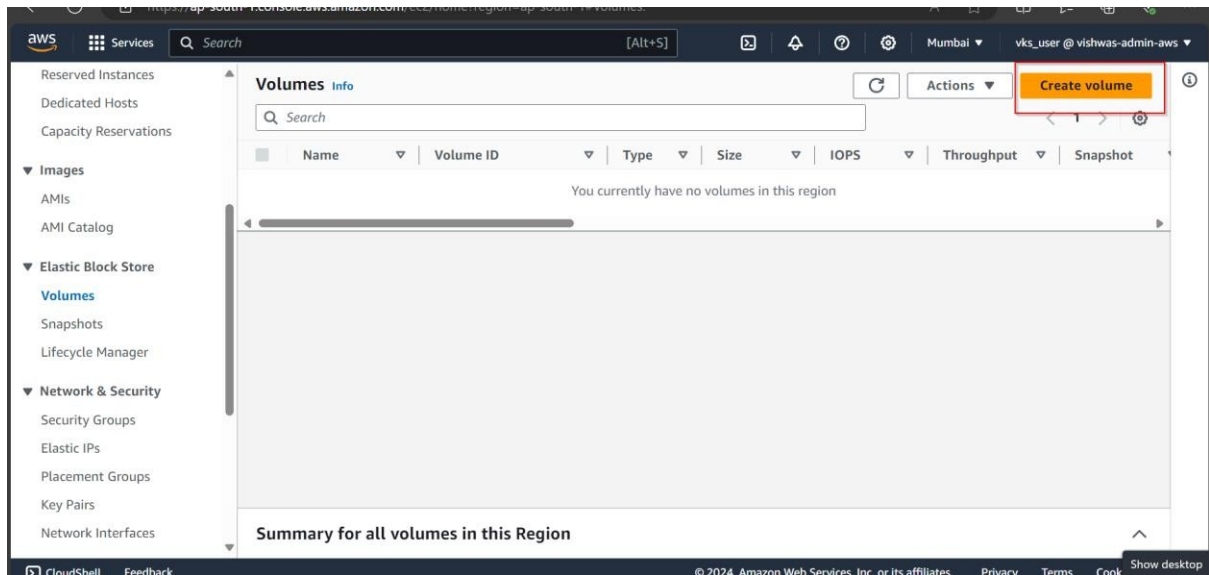
1.1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.



1.2. In the navigation pane, choose **Volumes**.



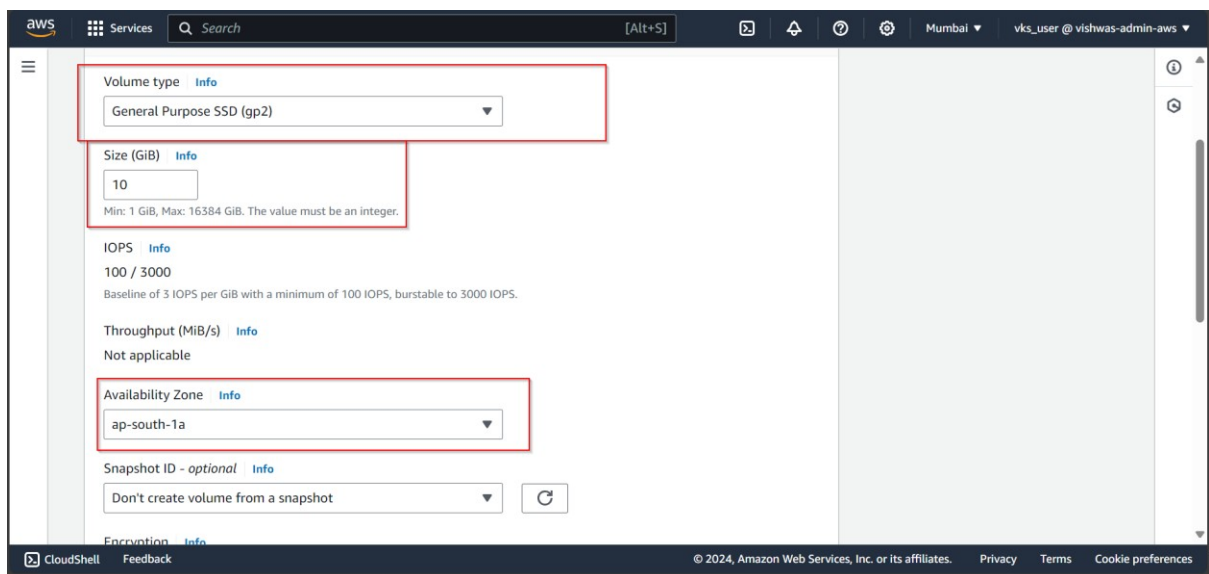
1.3. Click on **Create volume**.



1.4. For **Volume Type**, choose the type of storage you need. General Purpose SSD (gp2) is a good choice for a balance of price and performance.

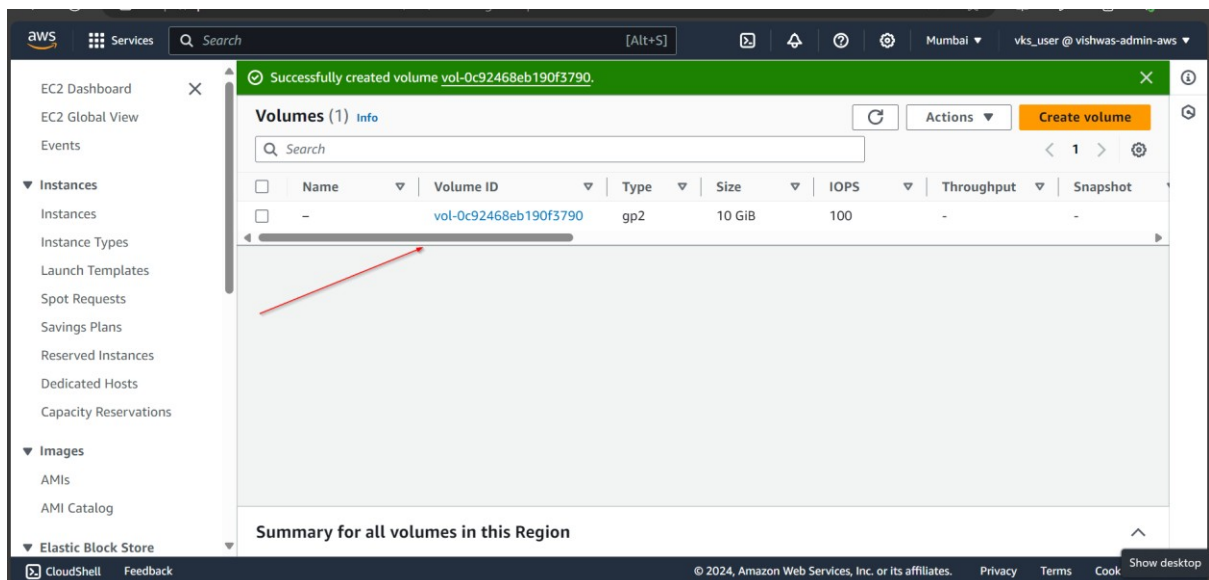
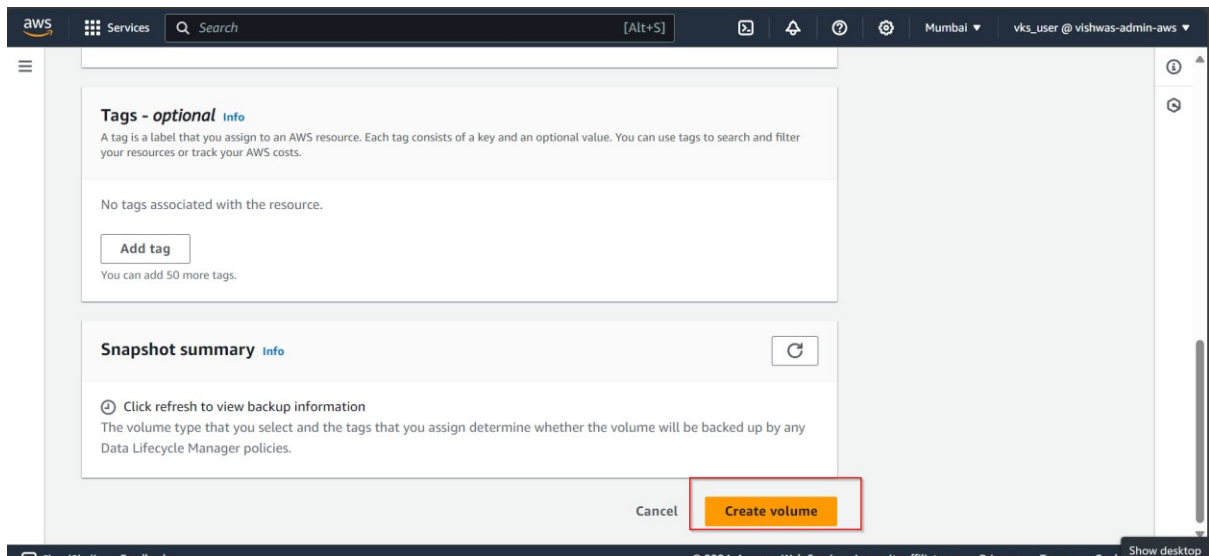
1.5. Enter the desired **Size** of the volume in GiB.

1.6. Choose the **Availability Zone** where you want the volume to be created.



1.7. (Optional) Add a **Tag** to the volume for easy identification. A tag is a key-value pair that you can assign to AWS resources.

1.8. Click **Create volume**.



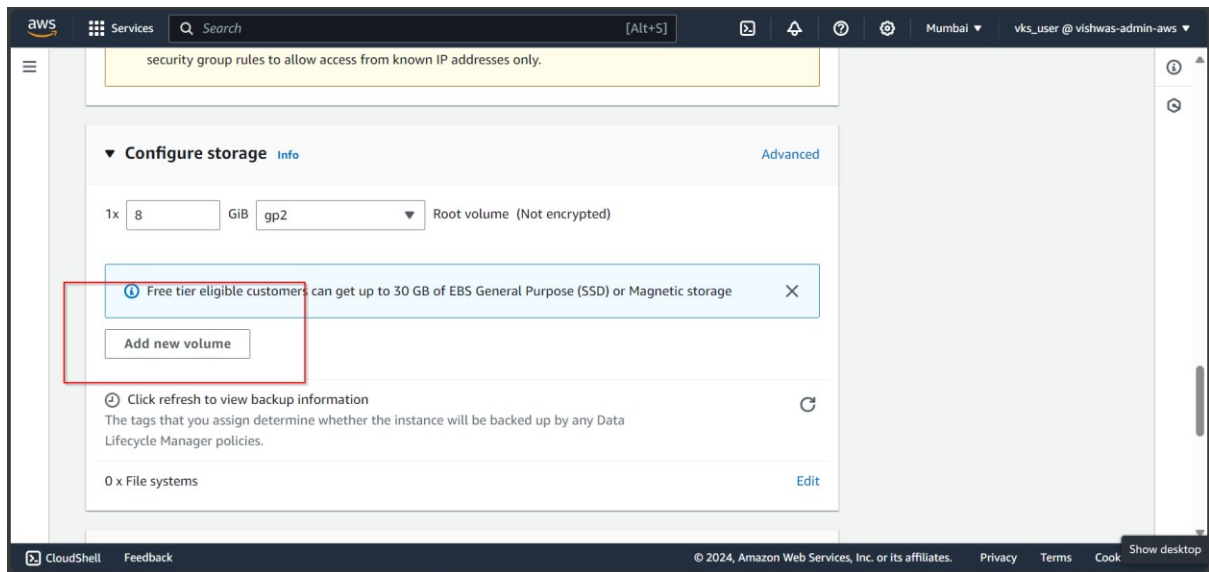
2. Attach the volume to an instance

2.1. In the navigation pane, choose **Instances**.

2.2. Select the instance that you want to attach the volume to.

2.3. Click on **Actions**, then select **Attach volume**.

You can attach a volume during creation or after creation

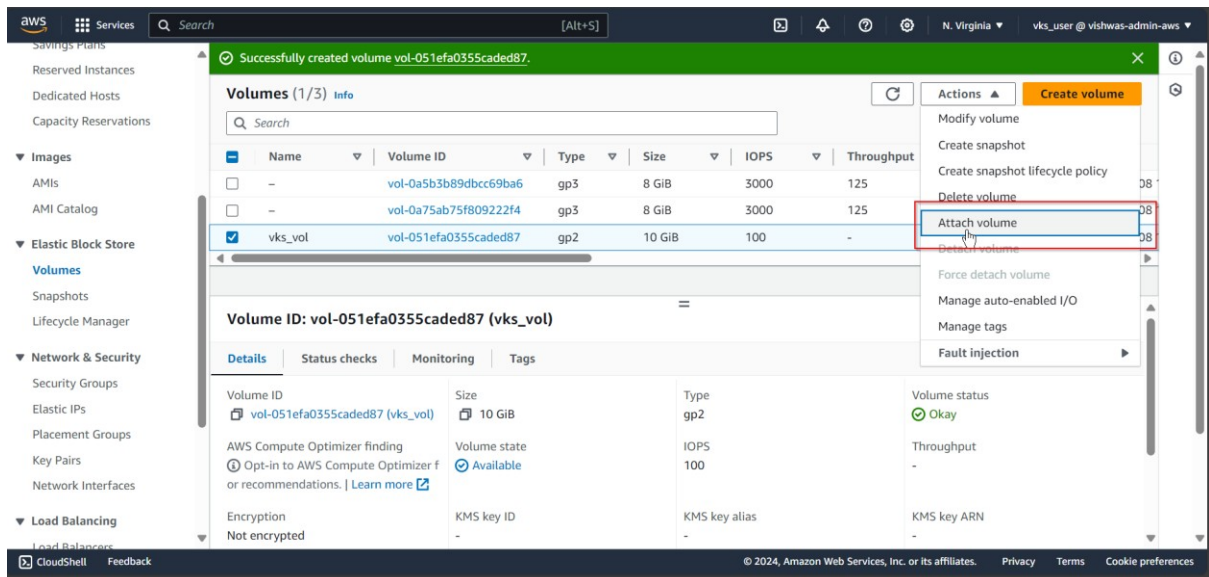


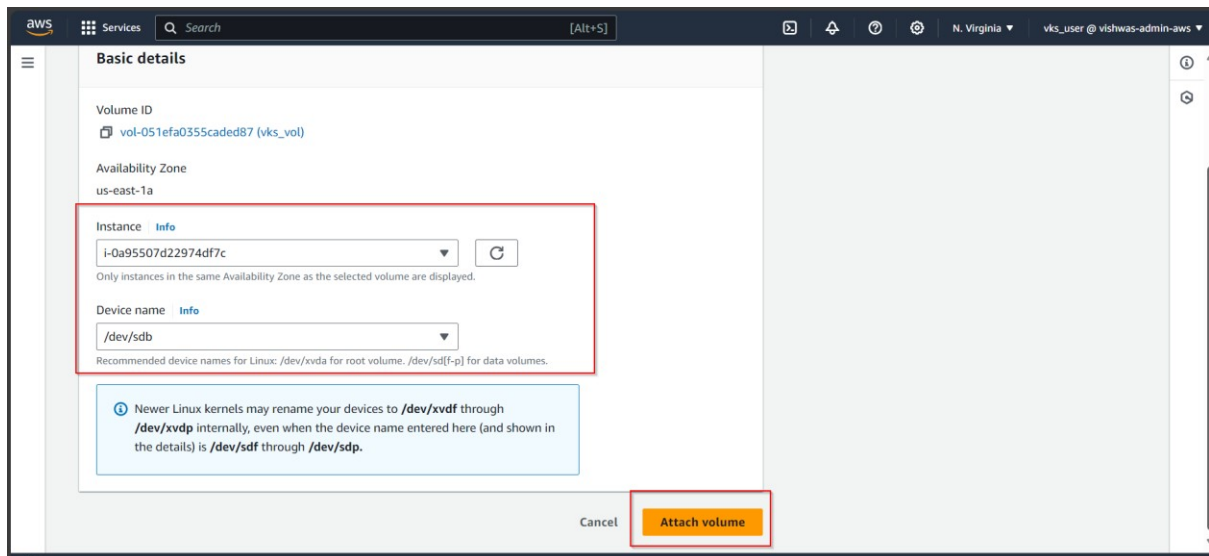
2.4. Under **Resources**, select the volume you created in step 1.

2.5. Under **Instance**, select the instance you want to attach the volume to.

2.6. In the **Device** field, select an available device name. You can choose from the recommended options or specify a custom device name.

2.7. Click **Attach**.

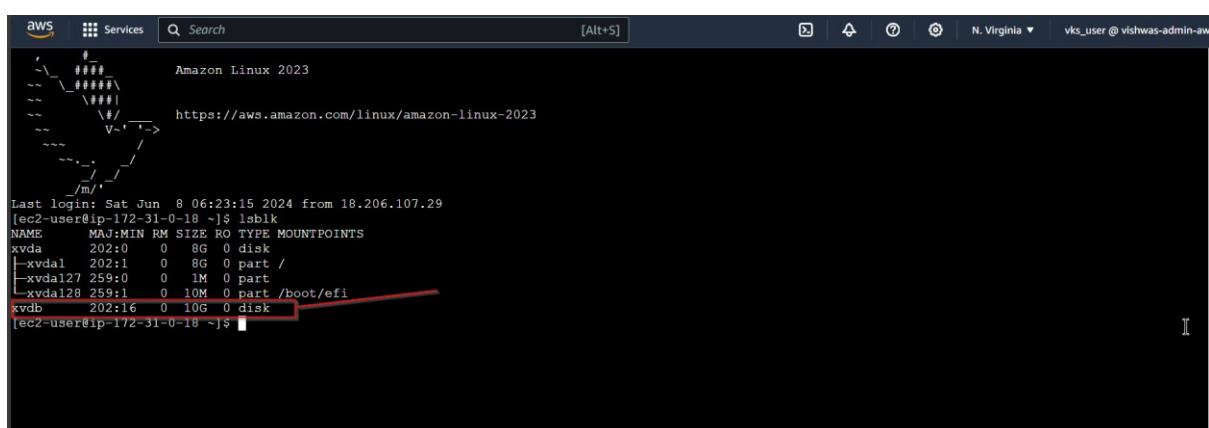




3. Configure the instance to use the virtual disk

3.1. Connect to your instance using SSH.

3.2. Once connected, you will need to use commands to format the volume and mount it to the file system. The specific commands will vary depending on your operating system. Here are some resources that you can refer to for formatting and mounting an EBS volume on Linux <https://docs.aws.amazon.com/ebs/latest/userguide/ebs-using-volumes.html>



```
aws Services Search [Alt+S] N. Virginia vks_user @ vishwas-admin-aw

https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sat Jun 8 06:23:15 2024 from 18.206.107.29
[ec2-user@ip-172-31-0-18 ~]$ lsblk
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0  8G  0 disk 
├─xvda1     202:1    0  8G  0 part /
├─xvda127   259:0    0  1M  0 part 
├─xvda128   259:1    0  10M 0 part /boot/efi
└─xvdb      202:16   0 10G  0 disk 
[ec2-user@ip-172-31-0-18 ~]$ sudo mkfs -t xfs /dev/xvdb
meta-data=/dev/xvdb          isize=512    agcount=4, agsize=655360 blks
                =              sectsz=512   attr=2, projid32bit=1
                =              crc=1        finobt=1, sparse=1, rmapbt=0
                =              reflink=1    bigtime=1 inobtcount=1
data        =              bsize=4096   blocks=2621440, imaxpct=25
                =              sunit=0     swidth=0 blks
naming      =version 2        bsize=4096   ascii-ci=0, ftype=1
log         =internal log    bsize=4096   blocks=16384, version=2
                =              sectsz=512   sunit=0 blks, lazy-count=1
realtime    =none            extsz=4096   blocks=0, rtextents=0
[ec2-user@ip-172-31-0-18 ~]$
```

```
naming      =version 2        bsize=4096   ascii-ci=0, ftype=1
log         =internal log    bsize=4096   blocks=16384, version=2
                =              sectsz=512   sunit=0 blks, lazy-count=1
realtime    =none            extsz=4096   blocks=0, rtextents=0
[ec2-user@ip-172-31-0-18 ~]$ sudo mkdir /data
[ec2-user@ip-172-31-0-18 ~]$ sudo mount /dev/xvdb /data
[ec2-user@ip-172-31-0-18 ~]$
```

```
aws Services Search [Alt+S] N. Virginia vks_user @ vishwas-admin-aw

[ec2-user@ip-172-31-0-18 ~]$ cd /data
[ec2-user@ip-172-31-0-18 data]$ sudo touch test_data.txt
[ec2-user@ip-172-31-0-18 data]$ sudo nano test_data.txt
[ec2-user@ip-172-31-0-18 data]$
```

```
aws Services Search [Alt+S] N. Virginia vks_user @ vishwas-admin-aw

GNU nano 5.8 test_data.txt Modified
I am test data

Help  Write Out  Where Is  Cut  Execute  Location  Undo  Set Mark  To Bracket
Exit  Read File  Replace  Paste  Justify  Go To Line  Redo  Copy  Where Was
```

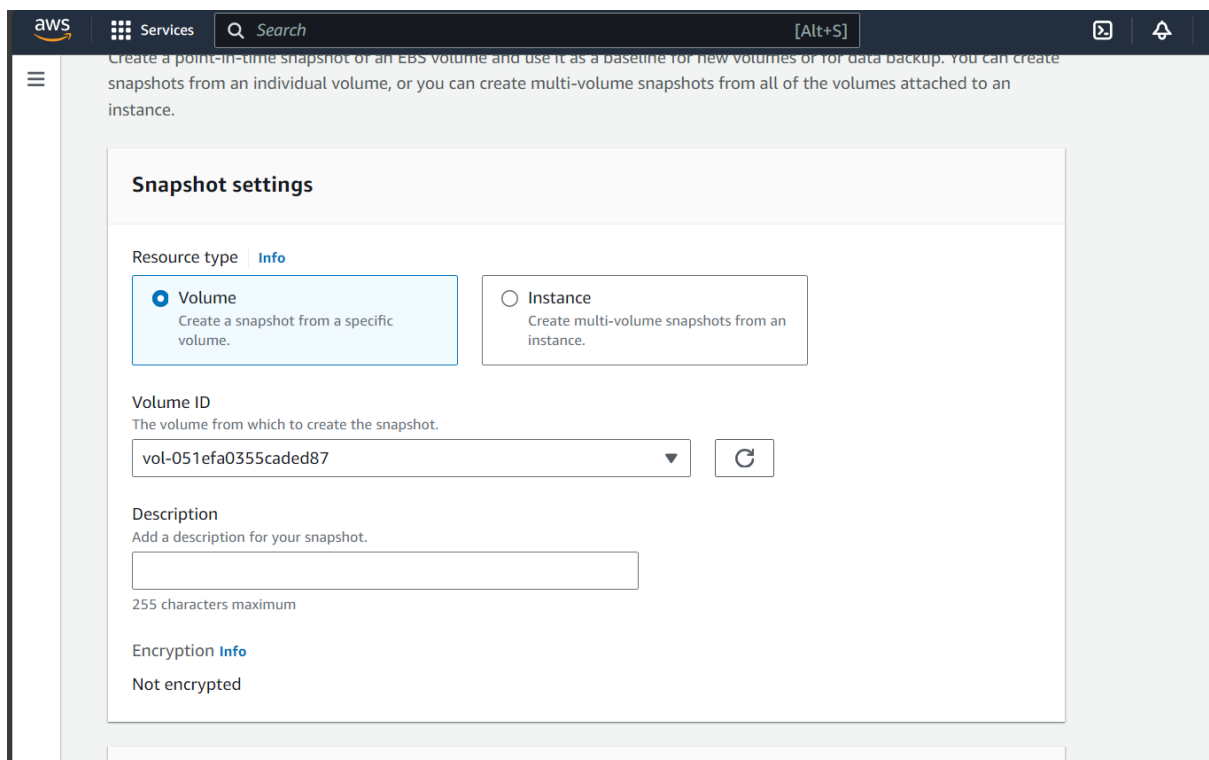
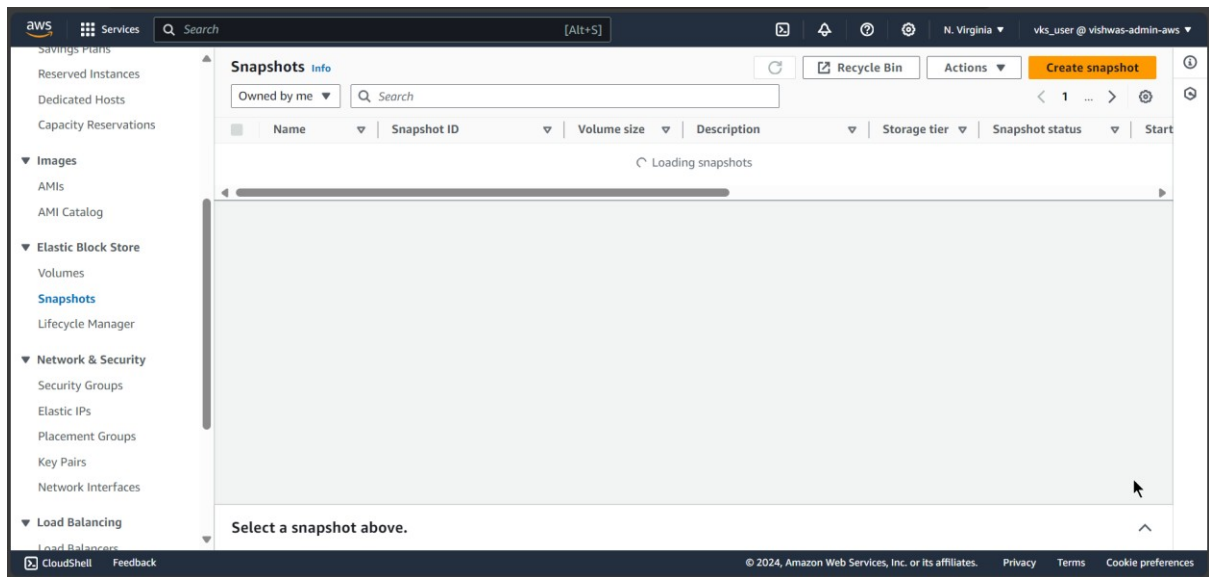
4. Create an Amazon EBS snapshot

Please Note

You can take a snapshot of an attached volume that is in use. However, snapshots only capture data that has been written to your Amazon EBS volume at the time the snapshot command is issued. This might exclude any data that has been cached by any applications or the operating system. If you can pause any file writes to the volume long enough to take a snapshot, your snapshot should be complete. However, if you can't pause all file writes to the volume, you should unmount the volume from within the instance, issue the snapshot command, and then remount the volume to ensure a consistent and complete snapshot. You can remount and use your volume while the snapshot status is `pending`.

- 4.1. Go back to the EC2 console and select **Volumes** in the navigation pane.
- 4.2. Select the volume that you want to create a snapshot of.
- 4.3. Click on **Actions**, then select **Create snapshot**.
- 4.4. Enter a **Description** for the snapshot.
- 4.5. Click **Create snapshot**.

```
[ec2-user@ip-172-31-0-18 data]$ less test_data.txt
[ec2-user@ip-172-31-0-18 data]$ cd
[ec2-user@ip-172-31-0-18 ~]$ sudo umount /data
[ec2-user@ip-172-31-0-18 ~]$
```



Volume ID
The volume from which to create the snapshot.

vol-051efa0355caded87

Description
Add a description for your snapshot.

255 characters maximum

Encryption [Info](#)
Not encrypted

Tags [Info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add tag](#)
You can add 50 more tags.

[Cancel](#) [Create snapshot](#)

Successfully created snapshot **snap-08df86fa3b3098fd0**.

Snapshots (1/8) [Info](#)

Owned by me

	Name	Snapshot ID	Volume size	Description	Storage tier	Snapshot status	Size
☐	xysfdgdshgs	snap-0fc59a18ba8cfaf10	10 GiB	jbsdnhsdghsfghsvfjsdajs...	Standard	Completed	20
☐	ath_snap	snap-03d1fa181caab75d3	10 GiB	Created TestData	Standard	Completed	20
☐	-	snap-0c91d916e520592f0	100 GiB	its aws snapshotttttt	Standard	Completed	20
☐	mg-ss	snap-0ed43da3f37b87fc2	21 GiB	mg volume snapshot	Standard	Completed	20
☐	-	snap-0dae70d0ee3dd09a3	10 GiB	-	Standard	Completed	20
☒	vks_snap	snap-08df86fa3b3098fd0	10 GiB	-	Standard	Pending	20
☐	tisha_snapshot	snap-0540d106a446f3ca2	10 GiB	tbfnrgdkjvkdjvdkvkn	Standard	Completed	20
☐	lol	snap-08c3664177feb5fcd	69 GiB	hehehehe-snapshot	Standard	Completed	20

Snapshot ID: snap-08df86fa3b3098fd0 (vks_snap)

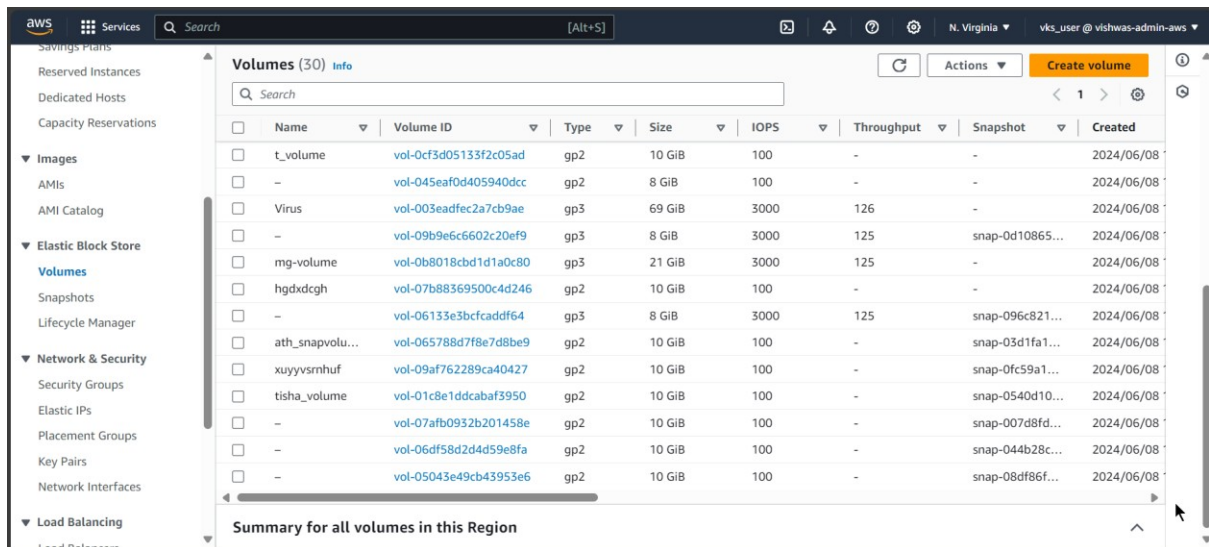
wait till completed

Details	Snapshot settings	Storage tier	Tags
Snapshot ID snap-08df86fa3b3098fd0 (vks_snap)	Volume size 10 GiB	Progress Unavailable	Snapshot status Pending

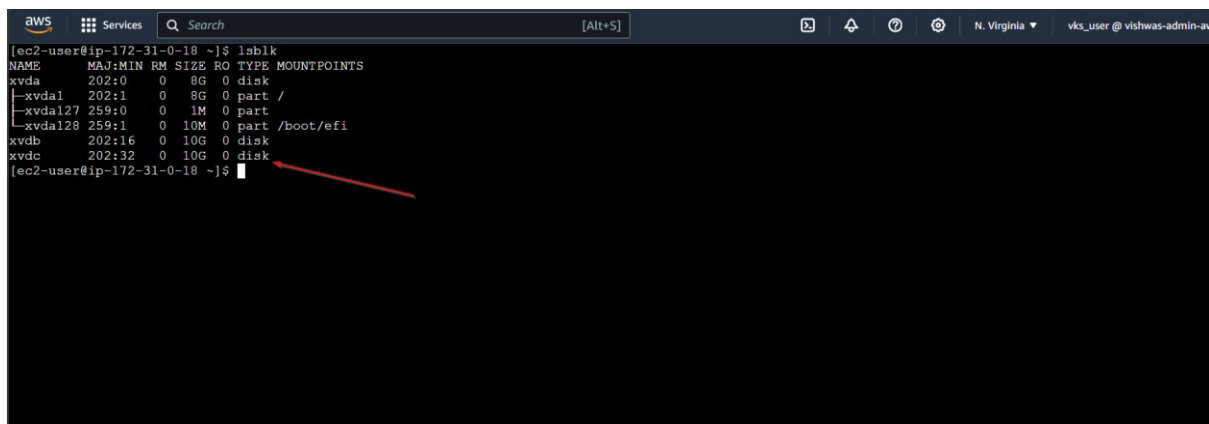
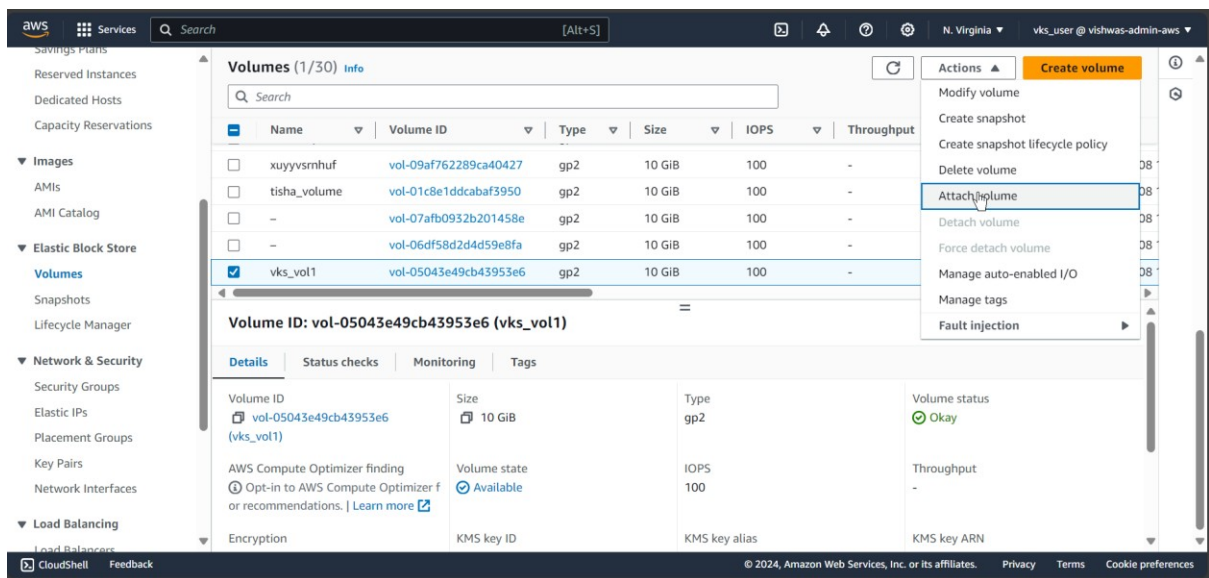
5. Restore the snapshot (Optional)

- 5.1. In the navigation pane, choose **Snapshots**.
- 5.2. Select the snapshot that you want to restore.
- 5.3. Click on **Actions**, then select **Create volume**.
- 5.4. Follow the steps in section 1.4 to create a new volume from the snapshot.

Attach the volume using the steps above (excluding formatting) and check



To find your volume – sort according to time and rename it



The screenshot shows an AWS console terminal window for an EC2 instance. The terminal output includes the following commands and results:

```
[ec2-user@ip-172-31-0-18 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk 
├─xvda1      202:1    0   8G  0 part /
├─xvda127    259:0    0   1M  0 part 
├─xvda128    259:1    0  10M  0 part /boot/efi
├─xvdb       202:16   0  10G  0 disk 
└─xvdc       202:32   0  10G  0 disk 

[ec2-user@ip-172-31-0-18 ~]$ sudo mkdir /data1
[ec2-user@ip-172-31-0-18 ~]$ sudo mount -t xfs /dev/xvdc /data1
[ec2-user@ip-172-31-0-18 ~]$ cd /data1
[ec2-user@ip-172-31-0-18 data1]$ ls
test_data.txt
[ec2-user@ip-172-31-0-18 data1]$ cat test_data.txt
this is test data
[ec2-user@ip-172-31-0-18 data1]$
```

A red box highlights the terminal output from the `lsblk` command and the subsequent mounting steps. A red arrow points from a text box labeled "Data in the back up volume" to the `test_data.txt` file listed in the directory.

Conclusion:

You have successfully created the backup and mounted the data in a new volume